

Riddhesh Dalal

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EDUCATION

GitHub: <https://github.com/artignite3>
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• **Indian Institute of Technology, Patna**

B. Tech, Chemical Science and Technology
• **Macro Vision Academy, Burhanpur**
Senior Secondary (XII) Board Examination (CBSE)
• **Macro Vision Academy, Burhanpur**
Secondary (X) Board Examination (CBSE)

CGPA: 8.04
July 2024 - Present
Percentage: 88.2%
June 2022 - May 2024
Percentage: 97.2%
July 2013 - May 2022

EXPERIENCE

• **Undergraduate Research Assistant at IIT Patna | Prof. Sriparna Saha**

IIT Patna, 2025 – Present

Machine Learning, Deep Learning & Generative AI

- Proposed Persistent Tactical Dynamics (PTD), a novel framework for modeling evolving team tactics using Hidden Markov Models and spatio-temporal player interactions.
- Implemented baseline models on SoccerNet-v3 benchmark (500 matches, 764 hours), integrating multi-object tracking with spatio-temporal graph neural networks for tactical impact analysis and game-state value modeling.
- Architected SM3 (TactiGraph: Tactical Player-Contribution), a player-contribution modeling system on SoccerNet video data using spatial Gaussian fields and pressure-sensitive ball-carrier metrics; research work currently being prepared for A/top-tier conference submission

• **Sub-Coordinator - Inter IIT Tech 14.0**

Sept. 2025 - Dec. 2025

Official Website & Security Application, Inter-IIT Tech Meet 14.0 **Source Code**

- **Tools & Technologies:** Next.js, React.js, Node.js, MongoDB, TypeScript, JWT, Vercel, React Native
- Developed and deployed the official full-stack website featuring multiple problem-statement, admin/contingent registration, and data-driven dashboards serving 23 IITs and supporting 2000+ contingent registration; hardened backend security with JWT authentication, and encrypted databases, Integrated a React Native companion app for real-time participant entry/exit management during the presentation.

PROJECTS

• **Redrob Intelligent Candidate Discovery Engine** (Team Lead, Team MM – 4 members)

June 2026 - July 2026

Python, BM25, FAISS, Sentence-Transformers, Cross-Encoder, Qwen2.5-LLM **Source Code**

- Built an explainable multi-stage AI candidate-ranking pipeline, ranking 100,000 candidates pool entirely CPU within a 5 minutes time limit for a hackathon.
- Designed a 7-stage AI candidate discovery pipeline, integrating BM25 lexical retrieval, embeddings(BGE-base-en-v1.5) with FAISS semantic search, behavioral signals, and neural re-ranking and natural-language rank explanations via Qwen 2.5-1.5B-Instruct
- Rectified a critical scoring bug by identifying an unflattened nested dictionary, restoring differentiated rankings across 100,000 candidate records.

• **AI-Driven Gamified Team Predictor - New X1**

Dec. 2025 - June 2026

Python, XGBoost, CatBoost, PuLP, FastAPI, Streamlit, SHAP, Ollama/Llama **Source Code**

- Engineered end-to-end ML pipeline predicting individual cricket player scores across 3000+ matches, 35000+ players records with ball-by-ball records and match conditions, then auto-generated optimal 11-player squads respecting role and composition limits.
- Benchmarked 2 Gradient boosting models, **XGBoost** and **CatBoost** regressors on 140 synthetic features for fantasy-point prediction; applied **PuLP** integer programming for team-selection optimization, with **SHAP**-based per-player explainability surfaced through a **FastAPI** + **Streamlit** application exposing real-time prediction APIs

• **Faculty Recruitment Portal** (Collaborative – Team of 2, with IIT Ropar)

Nov. 2025 - Feb. 2026

Node.js, MySQL, EJS, Bootstrap, JavaScript **Source Code**

- Delivered a full-stack recruitment portal with session-based authentication and role-based access for applicants/admins, backed by a normalized **MySQL** schema managing qualifications, employment history, and publications.

SKILLS/RELEVANT COURSEWORK

- **Languages:** Python, C/C++, JavaScript, TypeScript, SQL, HTML/CSS
- **ML/AI:** PyTorch, XGBoost, CatBoost, SHAP, Graph Neural Networks, Ollama/Llama, LangChain, LLM APIs
- **Web & Backend:** Node.js, FastAPI, Next.js, React.js, REST APIs, JWT
- **Databases & Storage:** MySQL, MongoDB, DuckDB
- **Tools & Platforms:** Git, GitHub, Linux, Vercel, Streamlit, React Native (Basic)
- **Key Courses:** Data Structures & Algorithms, Intro to Data Science, Probability & Statistics

ACHIEVEMENTS

• **300+** DSA Problems and **100+** SQL Queries solved (June 2026) • **Rank 13,641** (Top 5%) among 0.25 million candidates in JEE Advanced (May 2024) • **99.4%ile** – JEE Mains, out of 1.35 million candidates (Jan 2024) • **MAS, Top 1% (M.P.)** – IAPT NSEP Olympiad (Nov. 2023) • **Zonal Rank 1** – NSO, IMO (SOF Olympiad) (Mar. 2022)

POSITIONS OF RESPONSIBILITY

- **Overall Coordinator**, Vincetroke - Fine Arts Club, IIT Patna (May 2026 – Present) • **Production Lead**, TEDx IIT Patna (Apr. 2026 – Present) • **Contingent Member**, Inter-IIT Cult 8.0 (Dec. 2025) • **Dy. Cultural Minister**, Class 11th (June 2022) • **Head Boy**, Class 10th (Apr. 2021)